

A Mini Project-2 Report on
SHELF LIFE PREDICTION

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By

Srujuna Priya	(23251A6601)
Madhumita C	(23251A6638)
Khushi Goyal	(23251A6647)
Sreshtha T	(23251A6651)

under the guidance of
Mrs. P. Vaishnavi
Assistant Professor
CSE(AI&ML)



Department of Computer Science & Engineering (AI&ML)
G. Narayanamma Institute of Technology & Science
(Autonomous) (For Women)
Shaikpet, Hyderabad – 500104.

Affiliated to
Jawaharlal Nehru Technological University Hyderabad
Hyderabad - 500085
April, 2025

G. Narayanamma Institute of Technology & Science

(Autonomous)

(For Women)

Shaikpet, Hyderabad – 500104.

Department of Computer Science & Engineering (AI&ML)



Certificate

This is to certify that the Mini Project-1 report on “**SHELF LIFE PREDICTION**” bonafide work carried out by **Srujana Priya (23251A6601)**, **Madhumita C (23251A6638)**, **Khushi Goyal (23251A6647)**, **Sreshta T (23251A6651)** in the partial fulfillment for the award of B. Tech degree in Computer Science & Engineering (AI & ML), G. Narayanamma Institute of Technology & Science, Shaikpet, Hyderabad, affiliated to Jawaharlal Nehru Technological University, Hyderabad under our guidance and supervision. The results embodied in the Mini project work have not been submitted to any other University or Institute for the award of any degree or diploma.

Internal Guide

Mrs. P. Vaishnavi

Assistant Professor

Dept of CSE(AI&ML)

Head of the Department

Dr. O. Obulesu

Associate Professor&Head

Dept of CSE(AI&ML) and CSE(DS)

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Srujana Priya	(23251A6601)
Madhumita C	(23251A6638)
Khushi Goyal	(23251A6647)
Sreshtha T	(23251A6651)

ABSTRACT

Shelf life prediction of leafy vegetables is essential to ensure food safety, maintain nutritional quality, and reduce post-harvest losses. Leafy vegetables such as spinach are highly perishable due to their high moisture content and are prone to rapid deterioration caused by wilting, yellowing, texture changes, and microbial growth. Traditional methods of assessing freshness rely on manual inspection or laboratory testing, which are time-consuming, subjective, and often inefficient.

This project proposes an AI-based shelf life prediction system that uses image analysis to evaluate the freshness of leafy vegetables and estimate their remaining shelf life. The system analyzes visual features such as color variation, texture, and structural changes to classify the produce into different freshness stages - Fresh, Mid-life, and Near Spoilage. Based on this classification, the model estimates the remaining shelf life in days, providing a fast, non-destructive, and automated solution.

By integrating image preprocessing and machine learning techniques, the proposed system aims to support better storage decisions, reduce food wastage, and promote sustainable food management practices. Spinach is used as a prototype to develop and validate the model, demonstrating the practical applicability of AI in improving post-harvest quality assessment.

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1. INTRODUCTION

1.1 Motivation

Leafy vegetables are an essential part of a healthy diet, providing vital nutrients, vitamins, and minerals. However, they are also among the most perishable food products due to their high moisture content and delicate structure. Vegetables such as spinach deteriorate rapidly after harvest, leading to wilting, discoloration, texture degradation, and microbial spoilage. This short shelf life creates major challenges in storage, transportation, and retail management.

In many cases, freshness is assessed through manual inspection, which is subjective and inconsistent. Laboratory testing methods, although accurate, are time-consuming and not feasible for everyday use by farmers, vendors, or consumers. As a result, produce is often discarded prematurely or consumed when quality has already declined. This not only leads to economic losses but also contributes significantly to food wastage and environmental impact.

With the rapid growth of Artificial Intelligence and image processing technologies, there is an opportunity to develop an automated, non-destructive, and efficient system for freshness evaluation. The motivation behind this project is to create an AI-based shelf life prediction system that can analyze visual features of leafy vegetables and provide reliable freshness classification and remaining shelf life estimation. By offering a smart and practical solution, this project aims to reduce post-harvest losses, support better decision-making, and promote sustainable food management practices.

1.2 Existing Systems

Currently, shelf life prediction of leafy vegetables is primarily carried out using traditional and manual methods. One of the most common approaches is manual inspection, where freshness is judged based on visible characteristics such as leaf color, texture, firmness, and presence of wilting. However, this method is highly subjective, varies from person to person, and is not reliable for large-scale evaluation.

Another widely used approach is laboratory and chemical testing, which involves microbial analysis and chemical degradation studies. While these methods provide accurate results, they are time-consuming, expensive, and unsuitable for real-time or on-field assessment. They require specialized equipment and trained personnel, making them impractical for farmers, vendors, and retailers.

Sensor-based techniques are also used in some cases, where temperature and humidity levels are monitored to indirectly estimate shelf life. Although useful, these systems require additional hardware infrastructure and do not directly analyze the visual quality of the vegetables.

Traditional computer vision methods with basic machine learning techniques have also been explored. These methods rely on handcrafted features such as color thresholding and texture extraction. However, they lack robustness and often show lower accuracy compared to modern deep learning approaches.

Overall, existing systems are either subjective, expensive, indirect, or limited in accuracy, highlighting the need for an automated, non-destructive, and AI-based solution for effective shelf life prediction.

1.3 Literature Survey

Several research studies and technological advancements highlight the importance of using AI and image processing techniques for predicting the quality and shelf life of perishable produce:

Many researchers have explored computer vision methods to evaluate the freshness of vegetables and fruits by analyzing visual features like color, texture, and shape. Traditional machine learning models such as SVM, k-NN, and decision trees have been used to classify freshness levels based on handcrafted features extracted from images. Although these methods showed promising results, their performance was limited by the quality of feature extraction and sensitivity to lighting variations.

Recent studies show improved results with deep learning models, especially Convolutional Neural Networks (CNNs), which can automatically learn important visual features without manual intervention. CNN-based approaches have been used widely in food quality assessment, including prediction of ripeness, spoilage, and defect detection in fruits and vegetables. These models demonstrated higher accuracy and robustness compared to classical image processing techniques.

Research has also been conducted on shelf life estimation using both physical sensor data and visual analysis. For example, multispectral imaging and hyperspectral analysis have been used to monitor biochemical changes in produce, enabling more accurate shelf life prediction. However, these systems often require specialized and expensive equipment, limiting their practical usability.

Studies focusing on AI-based non-destructive methods highlight the potential of combining image analysis with machine learning to predict remaining shelf life in real time. These approaches are advantageous because they are faster, more cost-effective, and can be integrated with mobile or embedded systems for on-field use.

Overall, literature emphasizes that AI-based image analysis, particularly deep learning techniques, offers significant improvements in freshness classification and shelf life

prediction compared to traditional methods, motivating further development in this area.

1.4 Challenges to existing system

The existing methods for shelf life prediction of leafy vegetables face several limitations. Manual inspection, which is widely used, is highly subjective and depends on individual judgment. Different people may assess freshness differently, leading to inconsistent and unreliable results. This method is also not scalable for large quantities in markets or storage facilities.

Laboratory and chemical testing methods, although accurate, are time-consuming, expensive, and require specialized equipment and trained personnel. These techniques are not suitable for real-time evaluation or regular monitoring in farms, retail shops, or supply chains.

Sensor-based approaches that monitor environmental factors such as temperature and humidity provide only indirect estimation of freshness. They do not analyze the actual visual or structural condition of the leafy vegetables and require additional hardware infrastructure, increasing cost and complexity.

Traditional computer vision techniques using basic machine learning rely on handcrafted features like color thresholding and texture extraction. These methods often struggle with variations in lighting, background noise, and image quality, resulting in lower accuracy and poor generalization.

Overall, the key challenges in existing systems include subjectivity, high cost, lack of real-time capability, dependency on external hardware, and limited prediction accuracy. These limitations highlight the need for an automated, robust, and AI-based non-destructive solution for effective shelf life prediction.

1.5 Proposed System

The proposed system is an AI-based shelf life prediction model for leafy vegetables that uses image analysis to classify freshness stages and estimate the remaining shelf life. The system captures RGB images of leafy vegetables such as spinach, mint, coriander, and cabbage using a camera. These images are preprocessed through resizing, normalization, and augmentation to ensure consistent input quality.

A deep learning approach using Transfer Learning with MobileNetV2 is implemented to automatically extract important visual features such as color degradation, texture variation, and wilting patterns. The model classifies the vegetable into three freshness categories: Fresh, Mild (Mid-life), and Rotten (Near Spoilage). Based on the predicted category, the system estimates the remaining shelf life in days using a classification-to-time mapping approach.

Unlike traditional methods, the proposed system provides a fast, automated, and non-destructive evaluation of freshness without requiring laboratory testing or additional sensor infrastructure. It reduces human error, improves prediction accuracy, and enables real-time freshness assessment. This system aims to minimize post-harvest losses, support better storage and supply chain decisions, and promote sustainable food management practices.

1.6 Objectives

The main objectives of this project are:

- To design and develop a deep learning-based system for automated shelf life prediction of leafy vegetables using image processing techniques.
- To implement a Convolutional Neural Network (CNN) model based on MobileNetV2 for classifying freshness stages into Fresh, Mild (Mid-life), and Rotten.
- To perform image preprocessing techniques such as resizing, normalization, and augmentation to ensure consistent and reliable model input.

- To extract and learn important visual features like color degradation, texture changes, and wilting patterns from RGB images.
- To estimate the remaining shelf life (in days) based on the predicted freshness category using a classification-to-time mapping approach.
- To validate the system using spinach as a prototype and evaluate the model performance using accuracy metrics.

1.7 Methodology

The methodology of the proposed shelf life prediction system consists of several structured steps, starting from data collection to final prediction.

Initially, a dataset of leafy vegetable images (such as spinach, mint, coriander, and cabbage) was collected and categorized into three freshness classes: Fresh, Mild (Mid- life), and Rotten. The dataset was then divided into training, validation, and testing sets to ensure proper model evaluation.

In the preprocessing stage, all images were resized to 224×224 pixels to match the input requirements of the model. Normalization was applied to scale pixel values, and data augmentation techniques such as rotation, flipping, and zooming were used to improve model generalization and reduce overfitting. For model development, Transfer Learning was applied using a pre-trained MobileNetV2 model. The base model (trained on ImageNet) was frozen to retain learned features, and additional layers such as Global Average Pooling, Dropout (0.5), and a Dense output layer with Softmax activation were added for multi-class classification.

The model was trained using the Adam optimizer and categorical cross-entropy loss function. Early stopping was implemented to prevent overfitting and improve training efficiency.

Finally, the trained model predicts the freshness category of a given image along with a confidence percentage. Based on the predicted category, the system estimates the remaining shelf life in days using a predefined classification-to-time mapping approach.

1.8 Hardware & Software requirements

Hardware Requirements: Computer/Laptop with minimum 8GB RAM, Processor: Intel i5 or higher (or equivalent), Camera or Smartphone camera for image capture, Optional GPU (recommended for faster model training), Stable Internet connection (for research and optional cloud deployment).

Software Requirements: Operating System (Windows/Linux/macOS),

Programming Language: Python,

Libraries and Frameworks: OpenCV – for image processing,

NumPy and Pandas – for data handling,

Matplotlib / Seaborn – for visualization, Scikit-learn – for evaluation metrics,

TensorFlow / Keras or PyTorch – for deep learning model implementation,

Development Tools: Jupyter Notebook / VS Code / Google Colab, Dataset of categorized leafy vegetable images.

1.9 Organization of the project

The Shelf Life Prediction project is organized into multiple stages, ensuring a systematic development process from data collection to final evaluation.

The first stage involves data collection and preparation, where images of selected leafy vegetables are gathered and categorized into freshness levels. This is followed by the preprocessing stage, which includes resizing, normalization, and augmentation of images to ensure consistent input for the model.

The next stage focuses on model development, where Transfer Learning using MobileNetV2 is implemented. The architecture includes feature extraction, classification layers, and dropout for regularization. The model is trained, validated, and tested to evaluate its performance.

After training, the prediction and shelf life estimation module classifies the freshness stage and maps it to the estimated remaining shelf life in days.

Finally, the project includes testing, evaluation, and result analysis, where model accuracy and performance metrics are assessed. The system is structured in a way that allows future enhancements such as multi-vegetable classification and IoT integration for smart storage monitoring.

2. PROPOSED SYSTEM

2.1 Architecture

The architecture of the proposed Shelf Life Prediction system is based on a deep learning model using Transfer Learning with the MobileNetV2 convolutional neural network. The system processes input images of leafy vegetables and classifies them into different freshness stages.

Initially, the system accepts an input image of size $224 \times 224 \times 3$, which represents an RGB image of the leafy vegetable. The image is then passed to the MobileNetV2 model, which is pre-trained on the ImageNet dataset. In this architecture, the base layers of MobileNetV2 are kept frozen so that the previously learned features such as edges, textures, and color patterns can be effectively utilized for feature extraction.

The extracted feature maps are then passed through a Global Average Pooling 2D layer, which converts the high-dimensional feature maps into a one-dimensional feature vector. This helps reduce the number of parameters and simplifies the model while preserving important information.

Next, a Dropout layer with a dropout rate of 0.5 is applied. This layer randomly deactivates some neurons during training, which helps in reducing overfitting and improves the generalization capability of the model.

The processed features are then passed to a Dense layer consisting of three neurons. A Softmax activation function is used in this layer to generate probability values for each class.

Finally, the system classifies the input image into one of the three freshness categories: Fresh, Mild, or Rotten. Along with the predicted class, the system also provides a confidence score, indicating the probability of the prediction. This architecture enables efficient feature extraction, accurate classification, and reliable freshness stage detection for leafy vegetables.

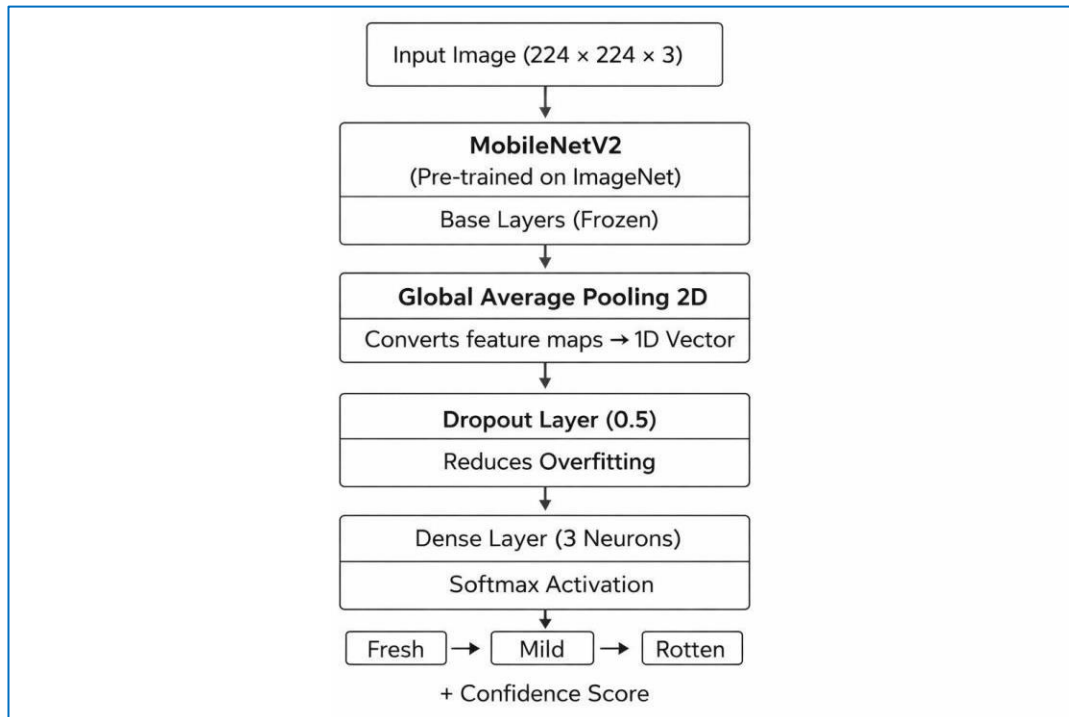


Figure 2.1.1 Architecture flow of the model

2.2 Module level design

The Shelf Life Prediction system is divided into several functional modules that work together to classify the freshness stage of leafy vegetables and estimate their remaining shelf life.

1. Image Input Module

This module is responsible for acquiring the input image of the leafy vegetable. The image is captured using a camera or selected from a dataset. The input image is resized to $224 \times 224 \times 3$ to match the input requirements of the deep learning model.

2. Feature Extraction Module

In this module, the input image is passed to the MobileNetV2 model, which is pre-trained on the ImageNet dataset. The base layers of the model are kept frozen so that previously learned features such as edges, color patterns, and textures can be effectively used to extract meaningful information from the image.

3. Feature Processing Module

The extracted feature maps are processed using a Global Average Pooling 2D layer, which converts the high-dimensional feature maps into a one-dimensional feature vector. This step reduces computational complexity while preserving the most important features required for classification.

4. Regularization Module

A Dropout layer with a rate of 0.5 is applied to the feature vector to reduce overfitting. This module randomly disables certain neurons during training, which helps improve the model's ability to generalize to new data.

5. Classification Module

The processed features are passed to a Dense layer with three neurons and a Softmax activation function. This module classifies the input image into one of the three freshness stages: Fresh, Mild, or Rotten.

6. Output Module

This module generates the final result by displaying the predicted freshness category along with the confidence score. Based on the predicted class, the system can further estimate the remaining shelf life of the leafy vegetable.

3. IMPLEMENTATION and TESTING

3.1 Implementation

The implementation of the Shelf Life Prediction system was carried out in a systematic manner, covering dataset preparation, preprocessing, model development, training, and prediction. The system was developed using Python along with deep learning frameworks such as TensorFlow/Keras.

3.1.1 Dataset Preparation

Images of selected leafy vegetables such as spinach, mint, coriander, and cabbage were collected and categorized into three freshness classes: Fresh, Mild (Mid-life), and Rotten. The dataset was organized into separate folders based on these categories. The complete dataset was divided into training and validation sets to enable effective model learning and performance monitoring.

3.1.2 Data Preprocessing

All images were resized to 224×224 pixels to match the input requirements of the MobileNetV2 model. Pixel values were normalized to improve training efficiency and convergence. Data augmentation techniques such as rotation, horizontal flipping, and zooming were applied to increase dataset diversity and enhance the robustness of the model.

3.1.3 Model Development

Transfer Learning was implemented using the pre-trained MobileNetV2 model with ImageNet weights. The base layers were frozen to preserve previously learned features. Additional layers were added, including a Global Average Pooling layer to reduce feature dimensions, a Dropout layer (0.5) to minimize overfitting, and a Dense output layer with Softmax activation consisting of three neurons representing Fresh, Mild, and Rotten categories.

3.1.4 Model Training

The model was compiled using the Adam optimizer and categorical cross-entropy loss function. Training was performed over multiple epochs while monitoring validation performance. Early stopping was applied to prevent overfitting and ensure optimal model performance.

3.1.5 Prediction and Shelf Life Estimation

After training, the model was used to classify new input images into one of the three freshness categories. The Softmax layer generated probability scores along with a confidence percentage. Based on the predicted freshness class, a classification-to-time mapping approach was applied to estimate the remaining shelf life in days, providing the final output of the system.

3.2 Testing

The testing phase was conducted to evaluate the performance and reliability of the proposed Shelf Life Prediction system. After training the model, it was tested using unseen images from the test dataset to assess its generalization capability.

During testing, input images were passed through the trained MobileNetV2-based model, and the predicted freshness category (Fresh, Mild, or Rotten) along with the confidence percentage was recorded. The predicted results were compared with the actual labels to measure classification accuracy.

Performance evaluation was carried out using accuracy as the primary metric. The model demonstrated satisfactory accuracy in classifying freshness stages of leafy vegetables, particularly spinach used as the prototype.

Additionally, the shelf life estimation module was verified by checking whether the predicted freshness class was correctly mapped to the corresponding remaining shelf life in days. The system successfully provided fast and consistent predictions, proving that image-based evaluation is more efficient compared to manual inspection methods.

Overall, the testing results confirm that the proposed AI-based system effectively classifies freshness stages and estimates shelf life in a reliable and automated manner.

3.3 Description of Technologies/Tools used

Frontend

Google Colab: Google Colab is an online notebook environment used to write and execute Python code. It provides an interactive interface where users can upload images, run the model, and visualize the output results. It also supports GPU acceleration which helps in faster model training and execution.

Backend

Python: Python is used as the main programming language for developing the system. It provides simple syntax and strong support for machine learning and image processing libraries.

TensorFlow / Keras: TensorFlow with Keras API is used for building, training, and evaluating the deep learning model. It provides high-level functions to implement neural networks efficiently.

MobileNetV2: MobileNetV2 is a lightweight Convolutional Neural Network (CNN) architecture used for feature extraction and image classification. It is pre-trained on the ImageNet dataset and helps in improving the model's accuracy while reducing computational cost.

OpenCV: OpenCV is used for image processing tasks such as reading images, resizing them, and performing basic preprocessing operations before feeding them into the model.

NumPy: NumPy is used for numerical computations and array manipulation. It helps in handling image data and performing mathematical operations during model processing.

Matplotlib: Matplotlib is used for visualization of images and training results such as accuracy and loss graphs during model development.

3.4 Modules Description

1. Image Input Module

Image Upload: Allows the user to upload an image of the leafy vegetable through the Google Colab interface.

Input Handling: The uploaded image is read and prepared for further processing by the system.

2. Image Preprocessing Module

Image Resizing: The uploaded image is resized to **224 × 224 pixels** to match the input size required by the MobileNetV2 model.

Normalization: Pixel values are scaled to ensure consistent input for the deep learning model.

Data Preparation: The processed image is converted into the appropriate array format for model prediction.

3. Feature Extraction Module

MobileNetV2 Model: A pre-trained **MobileNetV2** convolutional neural network is used to extract important visual features from the image.

Feature Detection: The model identifies patterns such as **color changes, texture variations, and wilting signs** in leafy vegetables.

4. Feature Processing Module

Global Average Pooling: Converts the extracted feature maps into a single feature vector to reduce model complexity.

Dropout Layer: Applies a **dropout rate of 0.5** to prevent overfitting and improve model generalization.

5. Classification Module

Dense Layer: The processed features are passed through a dense layer containing **three neurons**.

Softmax Activation: Generates probability scores for the three freshness categories **Fresh, Mild, and Rotten**.

6. Result Display Module

Prediction Output: Displays the predicted freshness stage of the leafy vegetable. **Confidence Score:** Shows the probability value indicating the confidence level of the prediction.

4. Results and Description

The proposed Shelf Life Prediction system was implemented using a deep learning approach with the MobileNetV2 model. The system was trained on a dataset of leafy vegetable images categorized into three classes: **Fresh, Mild (Mid-life), and Rotten**. The model was trained and executed using the **Google Colab** environment with the required machine learning libraries.

During training, the model learned to identify important visual features such as **color changes, texture variations, and wilting patterns** in leafy vegetables. The training process included image preprocessing, feature extraction using MobileNetV2, and classification through fully connected layers.

The trained model was able to successfully classify the freshness stage of leafy vegetables based on the input image. When a new image was uploaded, the system analyzed the visual characteristics and predicted the corresponding freshness category along with a **confidence score**. The results demonstrated that the deep learning model could effectively distinguish between fresh, mid-life, and rotten vegetables.

The output generated by the system provides a **quick and non-destructive method** for evaluating the freshness of leafy vegetables. Compared to traditional manual inspection methods, the proposed system offers **more consistent and automated results**. This helps in improving quality monitoring and can assist in reducing food wastage by identifying the remaining shelf life stage of leafy vegetables.

Overall, the results indicate that the proposed AI-based approach is effective for **freshness classification and shelf life prediction**, and it can be further improved with larger datasets and additional vegetable categories.

Conclusion

The Shelf Life Prediction System was successfully developed using deep learning and image processing techniques. The system uses a Convolutional Neural Network (CNN) based on the MobileNetV2 architecture to analyze images of leafy vegetables and classify them into different freshness stages such as Fresh, Mild, and Rotten.

The proposed system provides a fast and automated method for predicting the freshness level of vegetables without the need for manual inspection or laboratory testing. By analyzing visual features like color degradation, texture changes, and wilting patterns, the model can effectively determine the condition of vegetables and estimate their remaining shelf life.

The results demonstrate that image-based freshness detection is an efficient and reliable approach for food quality monitoring. This system can help reduce food wastage, support better storage management, and improve decision-making in food supply chains.

Future Work

Although the proposed system performs well in predicting vegetable freshness, there are several opportunities for future improvement and enhancement.

The dataset can be expanded with more images collected under different lighting and environmental conditions to improve model accuracy.

The system can be extended to support multiple types of vegetables instead of focusing on a single vegetable type.

Integration with mobile applications can enable users to capture images using smartphone cameras and receive instant freshness predictions.

The system can be integrated with IoT-based storage monitoring systems to track vegetable freshness in real-time.

Additional machine learning models can be explored to further improve prediction accuracy and system performance.

These future improvements can enhance the effectiveness and usability of the shelf life prediction system in real-world applications.

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Glossary

Artificial Intelligence (AI):

A branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence.

Machine Learning (ML):

A subset of artificial intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed.

Deep Learning:

An advanced type of machine learning that uses neural networks with multiple layers to analyze complex patterns in data such as images and videos.

Convolutional Neural Network (CNN):

A deep learning algorithm commonly used for image recognition and classification tasks by automatically extracting important features from images.

MobileNetV2:

A lightweight deep learning architecture designed for efficient image classification tasks, often used in mobile and embedded vision applications.

Image Processing:

A technique used to manipulate and analyze images in order to extract useful information or improve image quality.

Dataset:

A collection of images or data used to train and test machine learning models.

Image Preprocessing:

The process of preparing images before feeding them into a model, including resizing, normalization, and augmentation.

Freshness Classification:

The process of categorizing vegetables based on their condition such as fresh, mild, or rotten.

Shelf Life:

The length of time a product remains usable, fit for consumption, or retains its desired quality.